Fixing Shares That Are Not Really Halves or Fourths Answer Key

Kindergarten–Grade 1

Quick Answers

1. (a) the symbol = >, (b) squares in each true half = 4, —
2. (a) the symbol = >, (b) squares in each true fourth = 2, —
3. (a) the symbol = >, (b) squares in each true half = 6
4. (a) the symbol = >, (b) squares in each true fourth = 3, —
5. (a) the symbol = >, (b) squares in each true half = 4, —
6. (a) squares in each half = 6, (b) squares in each fourth = 3, —

What is verified

1 of 6 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 2, 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1
1.G – *problems 1, 2, 3, 4, 5, 6*
Difficulty 1–4 of 5 across 17 tagged checks.

Common wrong answers

1. *If they got 2: shared the strip into four parts instead of two*
2. *If they got 4: shared the strip into two parts instead of four*
3. *If they got 3: shared the strip into four parts instead of two*
4. *If they got 6: shared the strip into two parts instead of four*
5. *If they got 2: shared the rectangle into four parts instead of two*

1. Jonah cut a strip of eight squares into two parts, A and B, and called them halves. (a) Compare the parts. (b) How many squares would each true half have? (c) Explain why they are not halves.

Solution (a): Count along the strip. Part A covers five squares and part B covers three, so A holds more. $\boxed{5 > 3}$

Solution (b): The whole strip is eight squares. Two halves share the whole strip equally, so each half gets the strip shared between two: $8 \div 2$. $\boxed{4}$

Solution (c) — instructor-judged. A correct response ties "halves" to equal size. Full credit for the idea that halves must be the same size, and part A covers more squares than part B, so a cut anywhere but the middle does not make halves. Accept a spoken count (five against three) or a matching argument — laid on top of each other the parts do not line up. Saying only that the parts look different, without reference to the squares, earns half credit. Naming the bigger part alone earns no credit, since part (a) already asks for that.

2. Mira cut a strip of eight squares into four parts A, B, C, D and called them fourths. (a) Compare the biggest with the smallest. (b) How many squares would each true fourth have? (c) Explain why they are not fourths.

Solution (a): Part A covers three squares; part D covers one. Four pieces, but not four equal pieces. $\boxed{3 > 1}$

Solution (b): Share the eight squares among four equal parts: $8 \div 4$. $\boxed{2}$

Solution (c) — instructor-judged. A correct response says fourths must all be the same size. Full credit for the idea that cutting into four pieces is not enough — the four pieces have to match, and here they hold different numbers of squares. Accept a count of any two parts that differ. Believing that four pieces automatically means fourths, then correcting it on the spot with help, earns half credit. Naming the biggest part alone earns no credit, since part (a) already asks for that.

3. A strip of twelve squares was cut into two parts, A and B, and labelled halves. (a) Compare the parts. (b) How many squares should each half have?

Solution (a): Part A covers eight squares and part B covers four. $\boxed{8 > 4}$

Solution (b): Share the twelve squares between two equal parts: $12 \div 2$. The cut belongs two squares to the left of where it is drawn. $\boxed{6}$

4. Theo cut a strip of twelve squares into four parts A, B, C, D and labelled them fourths. (a) Compare part A with part D. (b) How many squares should each fourth have? (c) Explain which parts must change, and how.

Solution (a): Part A covers five squares and part D covers two. $\boxed{5 > 2}$

Solution (b): Share the twelve squares among four equal parts: $12 \div 4$. $\boxed{3}$

Solution (c) — instructor-judged. A correct response names what is too big and what is too small. Full credit for naming part A as too big and parts C and D as too small, together with the rule that every part has to end up holding the same number of squares as the others. Accept a description of sliding the cuts until each piece covers three squares. Naming only one wrong part earns half credit. Repeating the count of a true fourth earns no credit, since part (b) already asks for it.

5. Priya cut a rectangle of eight squares into two differently shaped parts, A and B, and called them halves. (a) Compare the parts. (b) How many squares should each half have? (c) Explain why differently shaped parts can still fail to be halves, and how to fix the cut.

Solution (a): Part A covers six squares — the whole top row plus two below it — and part B covers two. $\boxed{6 > 2}$

Solution (b): The rectangle holds eight squares, shared between two equal parts: $8 \div 2$. $\boxed{4}$

Solution (c) — instructor-judged. A correct response separates *shape* from *amount*. Full credit for the idea that halves are about how much is covered, not what the piece looks like, and that part A covers more squares than part B here, so the cut has to move until both parts cover the same number. Accept a described fix, such as cutting straight down the middle of the rectangle instead. Saying only that the pieces are different shapes earns half credit. Naming the bigger part alone earns no credit, since part (a) already asks for that.

6. Ella says the more pieces you cut, the bigger each piece is. One strip of twelve squares is shared into equal halves, another into equal fourths. (a) Squares in each half? (b) Squares in each fourth? (c) Explain whether Ella is right.

Solution (a): Twelve squares shared between two equal parts: $12 \div 2$. $\boxed{6}$

Solution (b): The same twelve squares shared among four equal parts: $12 \div 4$. $\boxed{3}$

Solution (c) — instructor-judged. Ella is wrong, and the two strips show it. Full credit for saying so with the reason that the same strip is being shared among more parts, so each part gets less — a fourth covers three squares while a half covers six. Accept the argument shown by pointing at the two strips rather than said in words. Saying she is wrong with no reason earns half credit. Agreeing with her earns no credit.